



UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/61

Paper 6 Probability & Statistics 1 (S1)

May/June 2012

1 hour 15 minutes

Additional Materials: Answer Booklet/Paper
 Graph Paper
 List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

The use of an electronic calculator is expected, where appropriate.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 50.

Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.

This document consists of **3** printed pages and **1** blank page.



- 1 It is given that $X \sim N(28.3, 4.5)$. Find the probability that a randomly chosen value of X lies between 25 and 30. [3]
- 2 Maria has 3 pre-set stations on her radio. When she switches her radio on, there is a probability of 0.3 that it will be set to station 1, a probability of 0.45 that it will be set to station 2 and a probability of 0.25 that it will be set to station 3. On station 1 the probability that the presenter is male is 0.1, on station 2 the probability that the presenter is male is 0.85 and on station 3 the probability that the presenter is male is p . When Maria switches on the radio, the probability that it is set to station 3 and the presenter is male is 0.075.
- (i) Show that the value of p is 0.3. [1]
- (ii) Given that Maria switches on and hears a male presenter, find the probability that the radio was set to station 2. [4]
- 3 A spinner has 5 sides, numbered 1, 2, 3, 4 and 5. When the spinner is spun, the score is the number of the side on which it lands. The score is denoted by the random variable X , which has the probability distribution shown in the table.

x	1	2	3	4	5
$P(X = x)$	0.3	0.15	$3p$	$2p$	0.05

- (i) Find the value of p . [1]
- A second spinner has 3 sides, numbered 1, 2 and 3. The score when this spinner is spun is denoted by the random variable Y . It is given that $P(Y = 1) = 0.3$, $P(Y = 2) = 0.5$ and $P(Y = 3) = 0.2$.
- (ii) Find the probability that, when both spinners are spun together,
- (a) the sum of the scores is 4, [3]
- (b) the product of the scores is less than 8. [3]
- 4 In a certain mountainous region in winter, the probability of more than 20 cm of snow falling on any particular day is 0.21.
- (i) Find the probability that, in any 7-day period in winter, fewer than 5 days have more than 20 cm of snow falling. [3]
- (ii) For 4 randomly chosen 7-day periods in winter, find the probability that exactly 3 of these periods will have at least 1 day with more than 20 cm of snow falling. [4]

- 5** The lengths of the diagonals in metres of the 9 most popular flat screen TVs and the 9 most popular conventional TVs are shown below.

Flat screen : 0.85 0.94 0.91 0.96 1.04 0.89 1.07 0.92 0.76
 Conventional : 0.69 0.65 0.85 0.77 0.74 0.67 0.71 0.86 0.75

- (i) Represent this information on a back-to-back stem-and-leaf diagram. [4]
- (ii) Find the median and the interquartile range of the lengths of the diagonals of the 9 conventional TVs. [3]
- (iii) Find the mean and standard deviation of the lengths of the diagonals of the 9 flat screen TVs. [2]
- 6** The lengths of body feathers of a particular species of bird are modelled by a normal distribution. A researcher measures the lengths of a random sample of 600 body feathers from birds of this species and finds that 63 are less than 6 cm long and 155 are more than 12 cm long.
- (i) Find estimates of the mean and standard deviation of the lengths of body feathers of birds of this species. [5]
- (ii) In a random sample of 1000 body feathers from birds of this species, how many would the researcher expect to find with lengths more than 1 standard deviation from the mean? [4]
- 7** (a) Seven friends together with their respective partners all meet up for a meal. To commemorate the occasion they arrange for a photograph to be taken of all 14 of them standing in a line.
- (i) How many different arrangements are there if each friend is standing next to his or her partner? [3]
- (ii) How many different arrangements are there if the 7 friends all stand together and the 7 partners all stand together? [2]
- (b) A group of 9 people consists of 2 boys, 3 girls and 4 adults. In how many ways can a team of 4 be chosen if
- (i) both boys are in the team, [1]
- (ii) the adults are either all in the team or all not in the team, [2]
- (iii) at least 2 girls are in the team? [2]

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